

ECON 403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Department of Economics
Nazarbayev University

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Lecture time/Location: 7e.329. Section 1L (403/503): 18:00–19:15, Section 2L: 19:30–20:45.

Office hours: By appointment, book weekly at <https://econ403-503.youcanbook.me>, please sign at least one day in advance.

Course Description This course analyzes problems and issues created by informational asymmetries and how properly designed market mechanisms could ameliorate those problems. Specifically, we will look at market mechanisms that can be classified as *auctions*. Auctions are currently used in a myriad of markets ranging from art to fresh fish, from electricity to government bonds, from government procurement to internet advertisements.

We will start the course by developing the tools required to analyze knowledge, beliefs, and information of economic agents. Then I will introduce the game theoretical model of *Bayesian Games*, which will be our main theoretical framework that we will use to model many auction formats.

After the introduction of the previous formal tools, we will examine the theoretical models behind the most commonly used auction formats. Students will learn the core theoretical results in auction theory.

Finally, we will introduce the basic theory of mechanism design in order to be able to discuss topics in auction design. Specifically, we will discuss the design of the optimal auction.

If time permits, I will discuss some advanced topics in auction theory, mechanism and information design.

Course Requirements: ECON 201 Intermediate Microeconomics (C- and above) and ECON 301 Econometrics I (C- and above); however, these are just the *minimum, official requirements*. I will assume you are proficient with the topics studied in Calculus, Probability, and Statistics, and that you are comfortable with analytical reasoning, mathematical notation, and proofs in order to be able to follow the lectures and readings.

Students should also have a solid and deep understanding of Microeconomic Theory at the intermediate/advanced level, with particular emphasis on topics such as utility maximization and choice under uncertainty. Specifically, students should have a solid understanding of Calculus, including differentiation, integration, the fundamental theorem of calculus, iterated integrals, and optimization. In Probability and Statistics, students should be familiar with the formal definition of random variables, probability distributions, expected value, conditional expectation, functions of random variables, and independence. Additionally, students should be comfortable with reading and understanding mathematical arguments and notations. They should also have some basic experience in constructing their own mathematical arguments and proofs. Therefore, it is strongly advised that you have a more extensive mathematical background than just a Calculus I course.

Furthermore, please note that this course has as an *unofficial prerequisite* knowledge of Game Theory at the level of ECON 302, or similar.¹ At the minimum, you should know what a normal-form game is, the concept of strategies, expected utility, dominance and rationalizability, Nash Equilibrium, mixed-strategy equilibrium, extensive-form games, and Subgame-Perfect Equilibrium.

Course Aims:

1. Understand the basic framework to model information and knowledge in strategic situations.
2. Analyze the main auction formats and the seminal results in auction theory.
3. Gain in-depth knowledge into the techniques used in modern microeconomic analysis.
4. Use the tools learned in the course to be able to model phenomena observed in real life.

¹ECON 305: Advanced Microeconomics is also suitable. There is a popular course on Coursera with similar contents that could serve as a substitute for ECON 302.

Course Learning Outcomes: By the end of the course the student will be expected to be able to

1. Demonstrate competence in using formal tools to model information in strategic situations.
2. Analyze games of incomplete information utilizing standard equilibrium concepts.
3. Demonstrate knowledge of the standard auction formats and contrast them in terms of their equilibrium outcomes.
4. Propose, construct and solve a simple mathematical model to analyze a phenomenon from real life using the tools from the course.

Learning and Teaching Methods: Lectures, problem sets, and a final essay.

Course Textbooks: There is no textbook required for this course. The course is based mainly on my lectures, slides and readings. All materials will be uploaded to Moodle.

The following is a list of undergraduate and graduate textbooks that you may find useful:

- Maschler, M., Solan, E., & Zamir, S. (2013). Game Theory. Cambridge: Cambridge University Press.
- Milgrom (2003), Putting Auction Theory to Work, Cambridge University Press.
- Krishna (2009), Auction Theory, Academic Press.

You will be responsible for all the material I cover or assign in class or tutorials. Therefore, the textbooks should be seen as a complement to the lectures are definitively *not a perfect substitute* for them.

Course Website: Additional course material, such as lecture slides, problem sets, and announcements will be posted on the Moodle page of the course, which you should check regularly.

Assessments

For ECO403 (UG students)	For ECO503 (MA students)
Problem Sets (15%)	Problem Sets (15%)
Essay (12%)	Research Proposal (22%)
Midterm (30%)	Midterm (25%)
Final exam (35%)	Final exam (30%)
Post-lecture Quizzes (5%)	Post-lecture Quizzes (5%)
Attendance (3%)	Attendance (3%)

Problem Sets We will have at least 5 problem sets. They will be assigned in class and will be due the next week after they are assigned. Late submissions will **not** be accepted. I will grade your problem sets on a scale from 0 to 2 where:

- 0: Not delivered.
- 1: Delivered but incomplete.
- 2: Delivered and complete.

We will discuss solutions to the problem sets in class or tutorials. *It is your responsibility to check your solutions against my class/tutorial discussion to see if you were correct or where you made mistakes.* Problem sets are difficult and are meant to represent the level of difficulty that you will face in the exam problems.

Essay Undergraduate students need to write a final essay for the course. The essay consists of picking a real life phenomenon and proposing a simple microeconomic theory model which models the situation. Students should be able to analyze the model and solve it and communicate their findings in written format in their essay. The correctness of the model, the quality and depth of the analysis and the solution, as well the quality of their writing will be assessed in the essay.

Research Proposal Master students have to work on a research proposal. The research proposal will have to be on a topic of your choice related to the general material of the course. You are welcome to talk to me to refine your topic. I will provide guidelines on how the research proposal should be written during the course.

Midterm The midterm will be on *Week 6*. It will cover roughly half of the material in the course. The exact date will be announced during the course.

Final Exam Will cover the second half of the material. The date will be chosen by the Registrar, during Final's Week.

Post-lecture Quizzes After each lecture, there will be a very short quiz deployed through Moodle, typically consisting of a single true/false question. These quizzes are designed to assess your understanding of the lecture content and encourage continuous study. The deadline for completing the post-lecture quizzes is 17:50 on the day of the next lecture.

Attendance All students are expected to attend the lectures and actively participate in class. If for some reason you are not able to attend to class, you should let me know in advance.

I will compute your final numeric grade by computing the weighted average of the aforementioned components according to the weights given in the table above.

Grading: Your course grade is determined by the weighted average of the assessments above according to the following scale:

A	95–100	A–	90–94.99
B+	85–89.99	B	80–84.99
B–	75–79.99	C+	70–74.99
C	65–69.99	C–	60–64.99
D+	55–59.99	D	50–54.99
F	0–44.99		

Course Expectations:

1. Class will begin promptly at the scheduled starting time. As a courtesy to others, it is requested that you not arrive after class has begun or leave before it ends. PLEASE NOTE: Since showing up late and leaving early is disruptive to the class, students who disrupt the class in this or any other manner may be administratively dropped or have their course grade reduced at the instructor's discretion. If you must leave class early or will be late, please let me know before class begins.
2. Class attendance will be taken regularly and randomly throughout the semester. If you fail to attend a lecture, you must submit the documents justifying your absence to the registrar and notify me by email. Students who have more than 25% of unexcused absences (7 lectures) will not be able to take the final exam. The instructor holds the right to excuse students from class or administratively drop them if necessary for violation of this policy.
3. *Exams* will be held outside of class, typically on Fridays or weekends. All exams are closed book and closed notes. The use of programmable calculators is prohibited. The dates for the midterm exams are fixed, but their locations will be announced later. The final exam will be scheduled by the Office of the Registrar.

If you miss an exam, you will receive a grade of zero unless you provide a valid and documented reason. In the case of a missed midterm exam with valid documentation, you will be allowed to take a make-up exam, which will be conducted as an oral exam. If you miss the final exam for a valid and documented reason, the University's Grading Policy regarding the grade of Incomplete will apply.

The final exam is cumulative. Appeals for midterm exam grades must be submitted in writing within five days of receiving the graded exam. Appeals for the final exam grade or the final course grade must also be submitted in writing within two days of the grades being posted.

Academic Misconduct: Students are expected to act with integrity. Cheating and plagiarism will not be tolerated. Any student deemed academically dishonest will be reported to the SSH disciplinary committee. Any cases of academic misconduct will be handled in accordance with the procedures of the Student Code of Conduct. The penalties may include anything from failing grade on a particular assignment to a loss of stipend, and even, dismissal from the University. You can read more about academic misconduct in the Student Code of Conduct. https://aao.nu.edu.kz/academic_integrity

AI Policy: This policy applies to any generative AI tool and all AI-generated products. You are discouraged from using AI tools unless explicitly instructed by me. If you are unsure or have questions, you must contact me before using AI for any assignment.

In this course, the potential risks of AI usage generally outweigh the benefits to student learning. If AI use is permitted, you are required to clearly indicate which parts of your assignment were created using AI and which parts are your original work. Additionally, no more than 25% of an assignment may be generated with AI, even when its use is approved.

You are required to sign an AI usage contract confirming your understanding and agreement to these policies. Unacknowledged use of AI tools will be treated as plagiarism, and penalties will be applied in accordance with the NU Student Code of Conduct.

Tentative Course Outline:

Session	Topics and Assignments	Course Aims	CLOs
Week 1	Introduction/Knowledge	1,3	1
Week 2	Knowledge/Beliefs	1,3	1
Week 3	Beliefs/Bayesian Games	1,3	1,2
Week 4	Bayesian Games	1,2,3,4	1,2
Week 5	Auctions with complete information	1,2,3,4	1,2,3
Week 6	Midterm 1		
Week 7	Bayesian Auctions, Private values	1,2,3	1,2,3
Week 8	Fall Break		
Week 9	Private values, Revenue Equivalence	1,2,3,4	1,2,3
Week 10	Extensions: asymmetries, risk aversion, reserve prices	1,2,3,4	1,2,3
Week 11	Introduction to Mechanism Design	1,2,3,4	1,2,3
Week 12	Optimal Auction Design	1,2,3,4	1,2,3
Week 13	Interdependent Value auctions	1,2,3,4	1,2,3
Week 14	Advanced Topics: revenue linkage, information	1,2,3,4	1,2,3
Week 15	Advanced Topics: Information Design	1,2,3,4	1,2,3

Disclaimer Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on Moodle.